



# **Catchment-Aware Flood Forecasting System**

R25 – 060

## **Project Proposal Report**

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**Component:**

Provide security guidance for flood-affected individuals  
to find safe sites and essential supplies.

## Declaration

I declare that this is my own work, and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or Institute of higher learning, and to the best of our knowledge and belief, it does not contain any material previously published or written by another person except where the acknowledgment is made in the text.

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**Date:** 24/01/2025

The above candidate is carrying out research for the undergraduate Dissertation under my supervision.

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## **Abstract**

Flooding is one of the most devastating natural disasters, causing significant damage to lives, property, and infrastructure. With the increasing prevalence of floods due to climate change and urbanization, there is a pressing need for advanced systems to enhance flood preparedness and response. This research focuses on developing a critical component of a Catchment-Aware Flood Forecasting System aimed at providing security guidance to flood-affected individuals. The solution incorporates predictive analytics to assess risks and issue timely warnings based on weather forecasts and flood monitoring data. Geospatial mapping and user location tracking enable the identification of safe sites and essential supplies, while explainable AI ensures transparent and interpretable navigation recommendations.

In addition to providing real-time updates on safe site statuses and supply availability, the system facilitates user communication and feedback to address immediate needs. Novel features include algorithms to dynamically suggest the nearest safe locations and supplies, alert users to changing flood conditions, and navigate them to safety using explainable AI. Literature indicates that integrating machine learning, GIS, IoT, and two-way communication enhances disaster management systems' efficacy. By combining these technologies, the proposed solution delivers a comprehensive, user-centric approach to mitigating flood risks and supporting affected populations. This system's adaptability to real-time conditions and focus on user safety and trust positions it as a significant advancement in flood disaster management practices, contributing to reduced loss of life and enhanced resilience in flood-prone regions.

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## **List of Abbreviations**

- XAI – Explainable Artificial Intelligence
- GIS – Geographic Information Systems
- SHAP -Shapley Additive Explanations
- LIME - Local Interpretable Model-agnostic Explanations
- NGO – Non-Government Organizations
- IoT – Internet of Things
- LTSM - Long short-term memory



## **Introduction**

The realm of the environment, infrastructure, and human lives are all greatly impacted by flooding, a devastating natural calamity. Advanced flood forecasting and management systems are required due to the growing frequency and intensity of floods brought on by urbanization and climate change [1]. Floods can be predicted, but effective flood management also involves giving impacted people timely, practical advice. In this regard, a Catchment-Aware Flood Forecasting System integrates contemporary technologies like artificial intelligence, geographic mapping, and predictive analytics to handle the complex issues of flood preparedness and response [2].

By directing flood victims to secure areas and giving them up-to-date information on necessary supplies, this system plays a crucial role in ensuring their safety and well-being. For mapping purposes, this component uses user position monitoring, predictive analytics to alert users to possible dangers, and algorithms to evaluate flood risks [3]. Incorporating explainable AI also enables the system to offer clear and understandable suggestions for directing users to secure websites, guaranteeing dependability and confidence in life-threatening circumstances. This method is unusual because it can adjust to changing flood conditions in real-time, giving users the most recent warnings and instructions while attending to their urgent requirements through user feedback and communication channels [4].

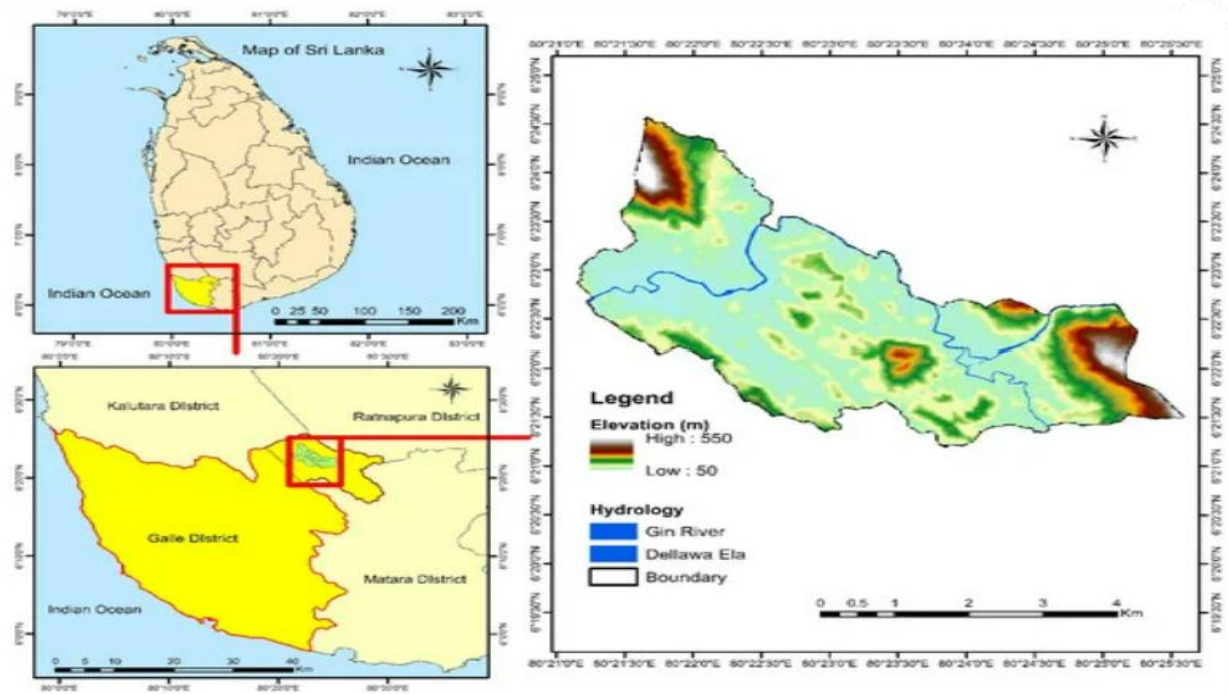


Figure 1 : Existing ways of displaying

## **Background and Literature Review**

Flood forecasting and management have been extensively studied as part of disaster risk reduction strategies. Traditional approaches primarily relied on hydrological models and rainfall-runoff analysis to predict flood occurrences. While these models provided a foundational understanding, they often lacked the precision and timeliness required for real-time decision-making [1]. The growing complexity of flood events, exacerbated by urbanization and climate change, underscores the need for more advanced systems capable of integrating diverse data sources such as weather forecasts, river gauge readings, and satellite imagery. Recent advancements in predictive analytics, machine learning, and artificial intelligence have transformed flood forecasting, offering innovative solutions to address these challenges [2].

### **Predictive Analytics in Flood Management**

Predictive analytics has emerged as a cornerstone for modern flood forecasting systems. Machine learning models, such as support vector machines, random forests, and neural networks, have been employed to analyze historical and real-time data to predict flood risks [3]. Studies like [2] highlight the superiority of ensemble models, which combine multiple algorithms to enhance prediction accuracy. Advanced deep learning techniques, such as Long Short-Term Memory (LSTM) networks, have shown remarkable potential in capturing the temporal dependencies inherent in hydrological data. [5] introduced an LSTM-based model integrated with explainable AI, achieving significant accuracy in water level prediction. The study highlighted how context-based preprocessing and explainable AI could enhance transparency and trust in flood forecasting systems, especially in regions like Sri Lanka where geographic complexities impact flood risks.

### **Geospatial Mapping and User Location Tracking**

Geospatial technologies are critical in visualizing flood-prone areas and guiding individuals to safety. The integration of Geographic Information Systems (GIS) with real-time data feeds allows for dynamic mapping of flood extents and safe zones [6]. Accurate user location tracking, combined with GIS-based flood mapping, has been shown to significantly reduce evacuation times in urban areas, ensuring that affected individuals can quickly navigate to safer locations.

### **Safe Site Identification and Resource Management**

Identifying and managing safe sites during flood events is crucial for the safety and well-being of affected populations. Recent studies emphasize the importance of crowdsourced data in monitoring resource availability at relief centers. For instance, [7] proposed a framework that leverages IoT sensors and mobile applications to provide real-time updates on essential supplies, including food, water, and medical resources. Furthermore, incorporating advanced predictive models, such as LSTM, allows for effective monitoring and allocation of resources in data-scarce regions [5].

### **Explainable AI in Disaster Management**

The application of explainable AI in disaster management is a relatively recent development, aimed at increasing transparency and trust in decision-making systems. Explainable AI techniques, such as SHAP (Shapley Additive Explanations) and LIME (Local Interpretable Model-agnostic Explanations), enable users to understand the rationale behind system-generated recommendations. [5] demonstrated the importance of identifying the contributions of individual catchment areas to flooding events using explainable AI. By enhancing user trust and compliance, explainable AI ensures that individuals are more likely to follow navigation and safety instructions, which is critical in high-stakes scenarios like floods.

### **User Communication and Feedback Mechanisms**

Effective communication is a cornerstone of disaster management. Mobile applications with two-way communication capabilities have been shown to significantly improve disaster response outcomes. [4] demonstrated the value of user feedback in refining evacuation routes and resource distribution. By incorporating real-time user communication and feedback mechanisms, systems can dynamically adapt to the evolving needs of flood-affected populations.

### **Conclusion**

The proposed solution to provide security guidance for flood-affected individuals aligns with recent advancements in flood forecasting and management systems. By integrating predictive analytics, geospatial mapping, and explainable AI, this component delivers a comprehensive and user-centric approach to flood risk mitigation. The inclusion of real-time updates, resource

monitoring, and user communication ensures that the system remains responsive to dynamic flood scenarios, ultimately enhancing its effectiveness in safeguarding lives and resources during flood events.

## **Research Problem**

Despite significant advancements in flood forecasting technologies, current systems often fall short in providing actionable and real-time guidance to individuals directly affected by floods. Many existing solutions focus on flood prediction and monitoring but lack comprehensive features for ensuring user safety, such as personalized navigation to safe sites, real-time updates on resource availability, and two-way communication for support. Furthermore, the absence of explainable AI in navigation systems limits user trust and understanding of recommendations, which is critical in high-stakes situations. Addressing these gaps is essential to develop a holistic and user-centric flood management solution that enhances safety, trust, and adaptability during flood events.

## **Main Objectives**

- To develop a Catchment-Aware Flood Forecasting System that integrates predictive analytics, geospatial mapping, and artificial intelligence which provides security guidance to flood-affected individuals, ensuring their safety and well-being during flood events.

## **Specific Objectives**

- To design and implement algorithms that assess flood risks and provide timely warnings based on weather forecasts and monitoring systems.
- To develop a user location tracking and mapping system for identifying and navigating safe sites.
- To enable real-time updates on safe site statuses, supply availability, and flood conditions.

- To incorporate XAI techniques to ensure transparent and interpretable recommendations for user navigation.
- To establish two-way communication mechanisms for users to provide feedback, request supplies, and receive support during flood events.
- To create novelty features, including dynamic suggestions for safe locations, alerts on flood risks, and real-time navigation to safety.

# Methodology

## 1. Problem Analysis and Requirement Gathering

- Conduct a comprehensive analysis of the challenges faced by flood-affected individuals.
- Identify key functional and non-functional requirements through literature reviews, stakeholder interviews, and case studies of existing flood management systems.
- Define the scope of the solution, focusing on real-time guidance, safe site identification, and resource availability.

## 2. Feasibility Study

### ❖ Technical Feasibility

- **Availability of Technology:**
  - The technologies required, such as predictive analytics, geospatial mapping, AI, and mobile app development (React Native or Flutter), are mature and widely accessible.
  - Integration of weather APIs, IoT devices, and GIS platforms is feasible with existing tools and frameworks.
- **Skills and Expertise:**
  - The project team possesses the required skills in data science, software development, and AI implementation.
  - Collaboration with meteorological and disaster management agencies can supplement technical knowledge.
- **System Scalability:**
  - Cloud-based platforms (e.g., AWS, Azure, or Google Cloud) can handle real-time data processing and support scalability during high-traffic events.

### ❖ Economic Feasibility



- **Cost Estimation:**
  - Initial development costs include software licenses, cloud infrastructure, and development team salaries.
  - Ongoing costs involve cloud services, data source subscriptions, and maintenance.
- **Funding Sources:**
  - Potential funding can come from government grants, disaster management agencies, or private-sector collaborations.
  - Sponsorships and partnerships with NGOs or environmental organizations may offset costs.
- **Cost-Benefit Analysis:**
  - The system can save lives and reduce property damage by providing timely guidance and justifying the investment.
  - Potential monetization through premium features (e.g., advanced alerts for businesses) can contribute to financial sustainability.

#### ❖ **Operational Feasibility**

- **Ease of Use:**
  - The system is designed with user-friendly interfaces to ensure accessibility for diverse user groups, including non-technical individuals.
  - Real-time updates and intuitive navigation features enhance usability during emergencies.
- **Stakeholder Support:**
  - Collaboration with local governments, NGOs, and community organizations ensures operational success.
  - Crowdsourced data collection from users fosters community engagement.

#### ❖ **Legal and Ethical Feasibility**

- **Data Privacy and Security:**

- Compliance with data protection regulations is essential for handling user location and personal data.
  - Secure data storage and transmission mechanisms will be implemented.
- **Ethical Considerations:**
  - Ensuring fair access to the system for all affected individuals, irrespective of socioeconomic status.
  - Transparent use of AI to build user trust in recommendations.

#### ❖ **Environmental Feasibility**

- **Impact on Environment:**
  - The system has no direct negative environmental impact and contributes positively by enhancing disaster preparedness and resilience.
  - Encouraging eco-friendly recovery and relief efforts through informed decision-making.

#### ❖ **Time Feasibility**

- **Project Timeline:**
  - The system can be developed and deployed within the project timeline by adopting agile development methodologies.
  - Phased development ensures critical features (e.g., flood alerts, and safe site identification) are prioritized for early deployment.

### **3. System Design and Architecture**

- **Architectural Framework:**
  - Design a modular system architecture incorporating predictive analytics, geospatial mapping, and AI algorithms.

- Develop a cloud-based infrastructure to enable scalability and real-time data processing.
- **Component Design:**
  - Risk analysis and predictive analytics module.
  - Geospatial mapping and location tracking module.
  - Safe site identification and real-time update mechanism.
  - User communication and feedback system.

#### **4. Data Collection and Integration**

- Gather data from multiple sources, including:
  - Weather forecasting systems (e.g., meteorological APIs).
  - Flood monitoring systems (e.g., river gauge data).
  - Geospatial data (e.g., maps, GPS, satellite imagery).
  - Crowdsourced information on safe sites and resource availability.
- Ensure the data is preprocessed for accuracy, consistency, and completeness.

#### **5. Algorithm Development**

- Flood Risk Assessment Algorithms:
  - Implement machine learning models (e.g., random forests, neural networks) to predict flood risks.
  - Use historical and real-time data for model training and validation.
- Explainable AI for Navigation:
  - Develop interpretable algorithms to suggest the safest routes and locations.
  - Use SHAP (Shapley Additive Explanations) or LIME (Local Interpretable Model-agnostic Explanations) to ensure transparency.
- Dynamic Safe Site Updates:
  - Create algorithms to monitor and update safe site statuses, including resource availability and crowd density.

## 6. System Development

- Front-End Development:
  - Build a user-friendly mobile and web applications for affected individuals and for the relevant authorities.
  - Provide real-time updates, alerts, and navigation interfaces.
- Back-End Development:
  - Develop APIs for predictive analytics, geospatial mapping, and communication.
  - Implement secure databases for storing user data, safe site information, and flood-related metrics.

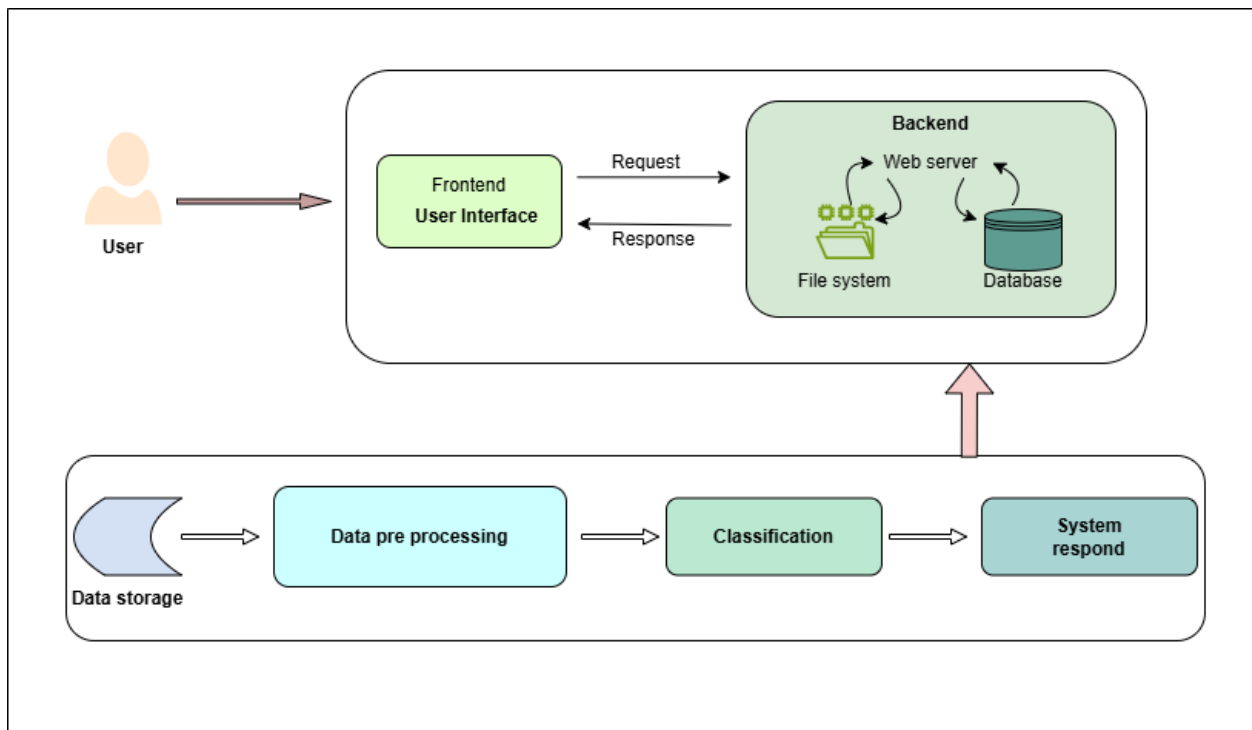


Figure 2 System Diagram

# **Project Requirements**

## **1.1 Functional Requirements**

### **1. Flood Risk Assessment**

- Develop algorithms to analyze flood risks using predictive analytics.
- Provide warnings based on weather forecasts and real-time monitoring data.

### **2. User Location Tracking and Mapping**

- Implement geospatial mapping to identify safe sites and routes.
- Enable user location tracking for personalized guidance.

### **3. Safe Site Identification**

- Identify and update the status of nearby safe locations dynamically.
- Monitor the availability of essential supplies at these sites.

### **4. Real-Time Alerts and Updates**

- Send alerts about nearby flood risks or changes in safe site statuses.
- Provide up-to-date information on supply availability and site overcrowding.

### **5. Navigation and Directions**

- Use explainable AI to navigate users to safe locations with clear and interpretable recommendations.

### **6. User Communication and Feedback**

- Facilitate two-way communication for users to request supplies or support.
- Collect user feedback to improve system functionality.

### **7. Novelty Features**

- Suggest the safest and nearest sites dynamically.
- Enable users to navigate safely with AI-driven decision-making.

## **1.2 Non-Functional Requirements**

### **1. Scalability**

- Ensure the system can handle a large number of users during disaster scenarios.

### **2. Reliability**

- Guarantee 24/7 uptime during critical periods like floods.

### **3. Performance**

- Deliver real-time alerts and updates with minimal latency.

### **4. Security**

- Protect user data and communication from unauthorized access.

### **5. Usability**

- Provide a user-friendly interface that is accessible to individuals with minimal technical expertise.

### **6. Interoperability**

- Integrate seamlessly with existing flood monitoring and weather forecasting systems.

### 1.3 Test Cases

- Perform unit, integration, and system testing to ensure functional accuracy.
- Conduct real-world simulations to test system performance under various flood scenarios.
- Validate the reliability and scalability of the system under high user loads.

#### Test case 1: The user finds the nearest safe site using the system

Table 1 - Test case 1

|                 |                                                                                                                                                                                                                                                                                                                  |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Test case ID    | 001                                                                                                                                                                                                                                                                                                              |
| Test case name  | The user finds the nearest safe site using the system                                                                                                                                                                                                                                                            |
| Description     | Verifies that the system correctly identifies and suggests the nearest safe site to the user based on their current location.                                                                                                                                                                                    |
| Pre-conditions  | The user must have location services enabled and internet access. The system must have updated safe site data.                                                                                                                                                                                                   |
| Testing steps   | <ol style="list-style-type: none"><li>1. Open the application</li><li>2. Allow location access if prompted.</li><li>3. Click on the "Find Nearest Safe Site" button.</li><li>4. Verify if the system displays the nearest available safe site.</li><li>5. Check if navigation directions are provided.</li></ol> |
| Expected output | The system should display the nearest safe site with navigation instructions and crowd status.                                                                                                                                                                                                                   |
| Actual output   | To be filled                                                                                                                                                                                                                                                                                                     |
| Status          | Pass/Fail (To be evaluated after testing)                                                                                                                                                                                                                                                                        |
| Remarks         | To be filled                                                                                                                                                                                                                                                                                                     |

## Test case 2: The affected user requests the essential dry rations using the system

Table 2 - Test case 2

|                 |                                                                                                                                                                                                                                                                                              |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Test case ID    | 002                                                                                                                                                                                                                                                                                          |
| Test case name  | The affected user requests the essential dry rations using the system                                                                                                                                                                                                                        |
| Description     | Test case verifies if users can successfully request essential dry rations using the system and receive confirmation.                                                                                                                                                                        |
| Pre-conditions  | The user must be logged into the system and connected to the internet. The system must have an updated inventory of available supplies.                                                                                                                                                      |
| Testing steps   | <ol style="list-style-type: none"><li>1. Open the application.</li><li>2. Navigate to the "Request Supplies" section.</li><li>3. Select the required dry rations from the available list.</li><li>4. Submit the request.</li><li>5. Verify if a confirmation message is displayed.</li></ol> |
| Expected output | The system should confirm the request and provide an estimated delivery/pick-up time.                                                                                                                                                                                                        |
| Actual output   | To be filled                                                                                                                                                                                                                                                                                 |
| Status          | Pass/Fail (To be evaluated after testing)                                                                                                                                                                                                                                                    |
| Remarks         | To be filled                                                                                                                                                                                                                                                                                 |



## **Commercialization and Business strategy**

The commercialization of the Catchment-Aware Flood Forecasting System focuses on delivering a subscription-based platform accessible via mobile and web applications. The product will provide features like real-time flood risk assessments, safe site navigation, and resource tracking, targeting local governments, disaster management agencies, and NGOs in flood-prone regions. A freemium model will ensure basic features are accessible to all users, while advanced analytics and reporting tools are offered as premium services. Enterprise solutions tailored for institutional clients will further enhance revenue generation.

The business strategy emphasizes building partnerships with meteorological departments, GIS providers, and NGOs to ensure data accuracy and system reliability. Initial market penetration will target high-risk districts in Sri Lanka, with plans to expand to other South Asian countries facing similar challenges. Marketing efforts will leverage social media, workshops, and collaborations with disaster management agencies to establish credibility and promote adoption. Continuous R&D and integration of emerging technologies, such as IoT sensors and satellite imagery, will ensure the system remains innovative and scalable for future applications.

## Timeline

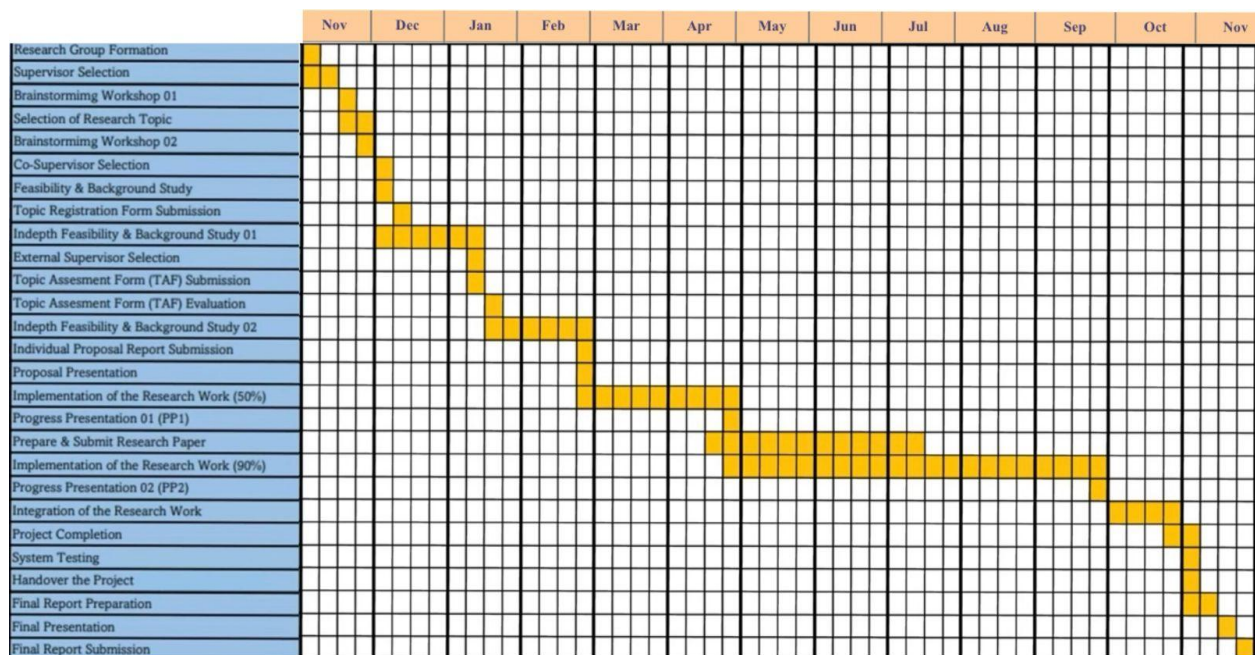


Figure 3 Gantt chart - Timeline

## Budget and Budget Justification

This project leverages technologies such as LSTM models, explainable AI, and geospatial mapping to address flood forecasting and management challenges.

| Item                                | Estimated cost |
|-------------------------------------|----------------|
| Data Acquisition and Licensing      | 2000           |
| Computational Resources             | 5000           |
| Software and Tools                  | 5000           |
| Development and Testing             | 7000           |
| Documentation and Publication       | 4000           |
| Miscellaneous Expenses (Additional) | 2000           |
| Total                               | 25000          |

## **Conclusion**

The integration of advanced technologies, such as predictive analytics, geospatial mapping, and explainable AI, offers a transformative approach to flood forecasting and management. By addressing key challenges like real-time decision-making, safe site identification, and user communication, the proposed Catchment-Aware Flood Forecasting System aims to enhance disaster preparedness and response. Leveraging machine learning techniques like LSTM and incorporating transparent AI solutions ensures both accuracy and user trust in life-critical scenarios. Furthermore, the inclusion of real-time updates and feedback mechanisms positions the system as a robust tool for safeguarding lives and resources during flood events. This research not only contributes to academic advancements in disaster management but also holds significant potential for practical applications in flood-prone regions like Sri Lanka, ultimately promoting resilience and reducing the devastating impacts of floods.

## References

- [1] W. M. O. (WMO), " Guidelines on Flood Forecasting and Warning Systems.," World Meteorological Organization (WMO), 2021.
- [2] J. & B. L. Smith, "Advances in Flood Forecasting and Disaster Management. Springer," *Advances in Flood Forecasting and Disaster Management. Springer*, 2020.
- [3] D. e. a. Chen, "Real-time Flood Forecasting Using Machine Learning Techniques," *Journal of Hydrology*, 588, 125036, 2020.
- [4] M. e. a. Lee, "Two-Way Communication in Disaster Management Applications," *International Journal of Emergency Management*, 14(3), 150-164, 2019.
- [5] Wijayaraja, J. L., Priyashantha, N., Gamage, N. D. U., Weerasekara, W. B. M. R., Pallegangoda, P. V. N. U., & Abeygunawardhana, P. K. W., "Understanding Catchment Impacts: Enhanced Water Level Prediction with LSTM and Explainable AI," *IEEE International Conference on Advances in Computing*, 2024.
- [6] Johnson, P., et al., "GIS Applications in Disaster Management," 2018.
- [7] A. e. a. Kumar, "IoT-enabled Resource Management for Flood Relief Operations," 2021.

